

ARPITA KUNDU

Water Resources Engineer | Hydro-Climate & Geospatial Machine Learning Researcher

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RESEARCH PROFILE

Undergraduate researcher in Water Resources Engineering with a strong quantitative foundation spanning surface-water hydrology, hydro-climate modelling, CMIP6-driven climate-impact assessment, flood susceptibility and inundation modelling, geospatial analytics, optical remote sensing, and data-driven environmental modelling. Bridges physically based hydrological-hydrodynamic simulation (HEC-HMS, HEC-RAS 2D) with machine-learning frameworks (Random Forest, XGBoost) to characterize river-basin dynamics, flood hazard, land-cover transformation, and climate-induced hydrological variability. Co-author on three conference manuscripts and one peer-reviewed journal article, with field-validated experience at Bangladesh's national water-management authority. Seeking a PhD position advancing computational hydrology, climate-resilient water-resources planning, and remote-sensing-machine-learning integration for flood-risk science.

EDUCATION

Bangladesh University of Engineering and Technology (BUET) **2022–2026**
B.Sc. in Water Resources Engineering — CGPA: 3.39/4.00 *Dhaka, Bangladesh*

- **Relevant coursework:** Open Channel Hydraulics, Hydrology, Groundwater Engineering, River Engineering, Flood Management, Numerical Methods, Remote Sensing & GIS, Engineering Statistics, Computational Hydraulics, Coastal Engineering.
- **Undergraduate thesis:** *Flood Susceptibility Mapping of the Surma River Basin Using Machine Learning and Multi-Source Remote Sensing* (Supervisor: Dr. Nasreen Jahan).

RESEARCH INTERESTS

Computational and physically based hydrological modelling | Flood susceptibility, hazard, and inundation mapping | Climate-change impact assessment in river basins (CMIP6 downscaling) | Remote sensing for environmental and hydro-climatic monitoring | Machine learning and explainable AI for hydrology | Urban heat island and land-surface-temperature dynamics | GIS-based water-resources and floodplain planning.

TECHNICAL SKILLS

Programming	Python (NumPy, pandas, scikit-learn, GeoPandas, rasterio, Matplotlib), MATLAB, Google Earth Engine (JavaScript API), basic SQL.
Hydrology & Hydraulics	HEC-HMS, HEC-RAS (1D/2D), SWMM, CMIP6 climate-scenario processing, rainfall-runoff and hydrodynamic flood-inundation modelling.
Geospatial & RS	ArcGIS, QGIS, Google Earth Engine; DEM-based terrain analysis, watershed delineation, NDVI/NDWI/LST retrieval, Landsat & Sentinel preprocessing, radiometric & atmospheric correction.
Machine Learning	Random Forest, XGBoost, logistic regression; feature engineering, hyperparameter tuning, cross-validation, data-leakage mitigation, ROC-AUC evaluation, spatial validation.
Engineering Design	AutoCAD, CSI SAP2000; structural detailing, reinforcement layout, drainage and culvert design.
Other	L ^A T _E X, Git/GitHub (version control), scientific writing, reproducible analytical workflows.

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher — Flood Susceptibility Mapping, Surma River Basin **2025–2026**
Dept. of Water Resources Engineering, BUET | Supervisor: Dr. Nasreen Jahan

- Built an end-to-end geospatial machine-learning pipeline integrating remote sensing, GIS, and ensemble classifiers to generate basin-scale flood susceptibility maps across the trans-boundary Surma River Basin.
- Automated extraction, mosaicking, and radiometric calibration of multi-temporal Landsat and Sentinel imagery in Google Earth Engine, enabling scalable, cloud-based ingestion of multi-source datasets and reducing manual preprocessing time substantially.
- Derived a suite of ~12 hydro-geomorphic conditioning factors (elevation, slope, curvature, TWI, SPI, drainage density, distance-to-river, rainfall, land use, soil, and lithology) from 30 m SRTM DEMs using ArcGIS and QGIS spatial-analysis workflows.
- Trained, tuned, and benchmarked Random Forest and XGBoost classifiers in Python, applying stratified train-test splitting, k-fold cross-validation, and ROC-AUC evaluation to identify the most predictive susceptibility model.
- Implemented disciplined feature engineering—multicollinearity screening, noise reduction, spatial-consistency checks, and explicit data-leakage mitigation—to strengthen generalization and produce a reproducible, decision-support-ready flood-risk framework.

Research Assistant — Urban Vegetation & Land-Surface-Temperature Dynamics **2023–2024**
Remote Sensing & Urban Climate Study, Dhaka District

- Designed multi-spectral Landsat processing workflows in ArcGIS—image preparation, radiometric correction, and

spatial transformation—to evaluate two decades of urban micro-climate change across Dhaka district.

- Computed NDVI and retrieved Land Surface Temperature to quantify vegetation degradation and urban thermal stress, then performed pixel-level spatial correlation linking declining vegetation density to intensifying urban heat-island conditions.
- Translated findings into an applied remote-sensing study connecting land-cover transformation, environmental monitoring, and climate-sensitive urban planning, published as a peer-reviewed journal article (see Publications).

PROFESSIONAL EXPERIENCE

Engineering Intern

Bangladesh Water Development Board (BWDB)

December 2025

Dhaka, Bangladesh

- Digitized and structured river cross-sections and layout plans in ArcGIS to support interpretation of riverbank-erosion patterns and river-reach morphology along active alluvial channels.
- Reviewed geometric profiles, engineering drawings, and material specifications for flood-control embankments, drainage regulators, and river-training structures.
- Plotted and interpreted surface-water hydrographs to characterize seasonal peak-flow behavior and prepare boundary conditions for hydraulic analysis.
- Processed raw hydro-climatological field observations into standardized technical sheets for regional water-resources planning, gaining hands-on exposure to institutional flood-protection workflows.

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed Journal Article

- **Kundu, A.,** Metu, M., Atique, M. M. H., Siam, N. S., & Sajid, S. (2024). Assessment of the change of vegetation cover and its impact on land surface temperature of Dhaka district using ArcGIS. *IUT Journal of Engineering and Technology (JET)*, 19(3), 33–43.

Conference Manuscripts (Under Review)

- **Kundu, A.,** et al. Climate-change impact on streamflow of the Khowai River Basin under CMIP6 scenarios using HEC-HMS. *International Conference on Advances in Civil Engineering (ICACE) 2026*. (Co-author, under review).
- **Kundu, A.,** et al. Evaluation of structural flood-mitigation strategies in the Sylhet region using HEC-RAS 2D hydrodynamic modelling. *International Conference on Advances in Civil Engineering (ICACE) 2026*. (Co-author, under review).

SELECTED RESEARCH PROJECTS

Climate-Change Impact on the Khowai River Basin

Co-Author — ICACE 2026 (Under Review)

- Downscaled and bias-corrected CMIP6 multi-model climate projections to drive future streamflow simulations and investigate long-term hydrological response across SSP scenarios.
- Constructed a calibrated HEC-HMS rainfall-runoff framework to quantify projected shifts in runoff generation, peak-flow timing, and climate-sensitive flow regimes.
- Delivered a scenario-based assessment workflow informing future water availability, flood potential, and climate-driven variability in basin-scale river response.

Flood Mitigation in the Sylhet Region

Co-Author — ICACE 2026 (Under Review)

- Evaluated competing mitigation alternatives—embankment positioning versus river dredging—to improve regional flood-management effectiveness following the 2022 Sylhet floods.
- Developed a high-resolution HEC-RAS 2D hydrodynamic model to simulate inundation extent, flow-velocity distribution, and spatial flood-propagation patterns.
- Coupled hydraulic outputs with planning considerations to support evidence-based appraisal of structural flood-control interventions.

Structural Design of a Multi-Cell Box Culvert

Academic Design Project

- Prepared detailed 2D structural layouts and reinforcement-spacing plans for a multi-cell box culvert sized from Hatirjheel drainage data and capacity requirements.
- Modelled culvert geometry and flow-passage configuration to enhance storm-water conveyance and mitigate localized waterlogging, integrating drainage hydrology with structural detailing and serviceability criteria.

HONORS & ACHIEVEMENTS

- **Top-8 Qualifier, University Innovation Hub Program** — Selected among the top eight teams for competitive problem-solving, innovation-oriented thinking, and communication of applied technical solutions at a university-level innovation platform.
- **Merit Admission to BUET** — Secured a place in Bangladesh’s most competitive engineering university through the national admission examination.

ADDITIONAL INFORMATION

Languages: English (professional working proficiency), Bengali (native), Hindi (conversational). **Referees:** Available on request, including thesis supervisor Dr. Nasreen Jahan (BUET).